# Ryan Algebra II Final Practice Grading

## Grading Note

The submitted reference image matches *Algebra II Final Targeted Practice*, not *Algebra II Final Topics Review*. I graded the visible work against the targeted practice sheet. Problems 5–12 were not visible in the provided reference image, so they are marked blank.

## Score Summary

Visible work score: 3.9/4 Overall submitted score: 3.9/12

| Problem | Credit | Note |
|---|---|---|
| 1 | 1/1 | Exact inverse-trig values are correct. |
| 2 | 1/1 | Correct factorization and solution set. |
| 3 | 1/1 | Correct cosine values and angles. |
| 4 | 0.9/1 | Features are essentially correct; amplitude should be positive 3. |
| 5–12 | 0/8 | No submitted work visible in the reference image. |

## Recreated Work and Corrections

1. Ryan's visible work:

$$\tan\theta = -\frac{5}{12}, \quad 5^2 + 12^2 = 169, \quad \sqrt{169} = 13, \quad \sec\theta = \frac{13}{12}$$

$$\sin\left(\cos^{-1}\left(-\frac{7}{25}\right)\right) = \frac{24}{25}, \qquad \cot\left(\sin^{-1}\left(\frac{4}{5}\right)\right) = \frac{3}{4}$$

Correct answer:

$$\boxed{\frac{13}{12}}, \qquad \boxed{\frac{24}{25}}, \qquad \boxed{\frac{3}{4}}$$

Credit: full.

2. Ryan's visible work:

$$2\cos^2\theta - \cos\theta - 1 = 0, \qquad (2\cos\theta + 1)(\cos\theta - 1) = 0$$

$$\cos\theta = -\frac{1}{2} \quad \text{or} \quad \cos\theta = 1$$

Correct answer:

$$\boxed{\theta = 0,\ \frac{2\pi}{3},\ \frac{4\pi}{3}}$$

Credit: full. Since $0 \le \theta < 2\pi$, $2\pi$ is not included.

3. Ryan's visible work:

$$\sqrt{3}\sec\theta + 2 = 0, \qquad \sqrt{3}\sec\theta = -2, \qquad \sec\theta = -\frac{2}{\sqrt{3}}$$

$$\cos\theta = -\frac{\sqrt{3}}{2}$$

   Correct answer:

$$\boxed{\theta = \frac{5\pi}{6}, \frac{7\pi}{6}}$$

   Credit: full.

4. Ryan's visible work:

$$y = -3\cos\left(2x + \frac{\pi}{2}\right) - 1 = -3\cos\left(2\left(x + \frac{\pi}{4}\right)\right) - 1$$

   Recreated features:

$$\text{amplitude written as } -3, \qquad \text{period } = \pi, \qquad \text{phase shift } = -\frac{\pi}{4},$$

$$\text{midline } y = -1, \qquad \text{range } [-4, 2]$$

   Correct answer:

$$\boxed{\text{amplitude } = 3}, \qquad \boxed{\text{period } = \pi}, \qquad \boxed{\text{phase shift } = -\frac{\pi}{4}}$$

$$\boxed{\text{midline } y = -1}, \qquad \boxed{\text{range } [-4, 2]}, \qquad \boxed{\text{one full period: } \left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]}$$

   Credit: nearly full. The vertical coefficient is $-3$, but amplitude is the positive distance 3.

5. No visible submitted work.

$$\boxed{y = 2\sin\left(\frac{1}{2}\left(x - \frac{\pi}{2}\right)\right) - 3}$$

6. No visible submitted work.

$$c = \sqrt{16^2 + 21^2 - 2(16)(21)\cos 58^\circ} \approx 18.5$$

$$K = \frac{1}{2}(16)(21)\sin 58^\circ \approx 142.5$$

$$\angle B \approx 75^\circ$$

   The larger angle is $\boxed{\angle B}$.

7. No visible submitted work. Since

$$\sin Q = \frac{24\sin 42^\circ}{18} \approx 0.892,$$

   two triangles are possible:

$$\boxed{Q \approx 63^\circ,\ R \approx 75^\circ,\ r \approx 26.0}$$

$$\boxed{Q \approx 117^\circ,\ R \approx 21^\circ,\ r \approx 9.7}$$

8. No visible submitted work. The bearings differ by $90^\circ$, so

$$d = \sqrt{5.2^2 + 7.1^2} \approx \boxed{8.8 \text{ miles}}.$$

9. No visible submitted work.

$$x^2 - 10x + 8y - 7 = 0$$

$$(x-5)^2 = -8(y-4)$$

This is a parabola with

$$\boxed{\text{vertex } (5,4)}, \quad \boxed{\text{focus } (5,2)}, \quad \boxed{\text{directrix } y=6}, \quad \boxed{\text{axis } x=5}.$$

10. No visible submitted work.

$$9x^2 + 4y^2 - 54x + 32y + 109 = 0$$

$$\frac{(x-3)^2}{4} + \frac{(y+4)^2}{9} = 1$$

This is an ellipse with

$$\boxed{\text{center } (3,-4)}, \quad \boxed{\text{major axis length } 6}, \quad \boxed{\text{minor axis length } 4}.$$

11. No visible submitted work.

$$25x^2 - 16y^2 - 200x - 32y - 16 = 0$$

$$\frac{(x-4)^2}{16} - \frac{(y+1)^2}{25} = 1$$

This is a hyperbola with

$$\boxed{\text{center } (4,-1)}, \quad \boxed{\text{vertices } (0,-1),(8,-1)}$$

$$\boxed{\text{foci } (4-\sqrt{41},-1),(4+\sqrt{41},-1)}, \quad \boxed{y+1 = \pm\frac{5}{4}(x-4)}.$$

12. No visible submitted work.

$$\boxed{\frac{(y-3)^2}{25} - \frac{(x+2)^2}{16} = 1}$$

Since $c^2 = a^2 + b^2 = 25 + 16 = 41$,

$$\boxed{\text{foci } (-2,3-\sqrt{41}),\ (-2,3+\sqrt{41})}.$$